

## **Course Project Abstract**

Connor Booth, Filip Droca

CP322: Machine Learning

Dr. Sukhjit Singh Sehra

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The online gaming platform Steam allows players to post reviews indicating whether or not they recommend a game. These reviews can provide a potential buyer with more information regarding others' satisfaction and engagement. For developers, understanding the factors that influence whether a player recommends a game can help them gain insight into what their game needs to succeed. This project will investigate whether review characteristics can predict whether a player will recommend a game.

The objective of this project is to build and evaluate machine learning models that predict whether a steam review is marked as “recommended” based on player behaviour and review characteristics. Player behaviours include total playtime, recent activity, and community engagement metrics. In addition, the project will explore whether text-derived features from the review can improve prediction accuracy or detect potentially misleading reviews. Such as a review which is sarcastic or has negative undertones, but is officially marked as positive.

The methodology involves several machine learning techniques commonly used for classification tasks. After preprocessing the dataset, Natural Language Processing (NLP) techniques will be applied to review the text to extract numerical features, such as review length and sentiment scores, which aim to capture the tone of the written review. These text-based features will be combined with behaviour variables, such as total play time. The data will then be split into train and test sets. Multiple classification models, such as Random Forest and Logistic Regression, will be trained, and performance will be compared using metrics like F1 score and AUC-ROC curves.

The dataset being used is the [Steam Reviews dataset](#) found on Kaggle. This dataset contains millions of user reviews and associated behavioural features. The dataset's size will allow our models to be trained on a large amount of real player feedback, increasing the reliability of the analysis.